Derelict Mill Alley — 15-Week Environment Art Plan

Updated Oct 12, 2026 · Dustin Apodaca

Overview

The goal is a 1:1 recreation of the reference: a derelict Victorian mill alley, about 40 m long, finished in 15 weeks (Mon Oct 19, 2026 to Sun Jan 31, 2027). It ships as a portfolio piece built the way a AAA team would build it: metrics-locked greybox, tiling materials and trim sheets, a reusable modular kit, a few unique hero assets, then dressing, lighting and renders.

What the piece must prove

- One kit, many walls: two building facades assembled from about 30 snapping modules on a 25 cm grid.
- Hero craft: the two-tier skybridge, the chimney silhouette and the basement light well break the modular rhythm and carry the shot.
- Material depth: vertex-painted height blends (brick, moss, grime, wet) and decals do the storytelling, not unique textures on every wall.
- A reference-matched camera: the final beauty shot lines up with the photo, plus 4 to 6 original angles.

Pipeline order

1. Pre-production: reference breakdown, scale, kit plan
2. Greybox: blockout, camera match, metrics lock
3. Materials: CC0 and procedural tiling sets, custom edge-wear and leak masks, trim sheets
4. Modular kit: meshes, UVs, snapping tests
5. Hero assets: Blender sculpt and retopo, Marmoset bake and texture
6. Props and debris
7. Foliage: ivy, moss, weeds, background trees
8. Set dressing: decals, vertex paint, storytelling
9. Lighting and atmosphere
10. Optimization and polish
11. Final renders and presentation

Assumptions

- Tools: Blender (modeling, sculpting, UVs, procedural material authoring, camera match, video edit), Marmoset Toolbag (baking, texturing, turntables, asset renders), Unreal Engine 5 (assembly, Lumen, Nanite, materials, set dressing, Movie Render Queue).
- Units: metric, 1 Unreal unit = 1 cm.
- Days: Monday, Tuesday, Friday, Saturday and Sunday, about one 4 to 6 hour session each; Wednesday and Thursday are off. The last workday of each week is review and catch-up. Christmas (Fri Dec 25) and New Year’s Day (Fri Jan 1) are buffer days, with five more buffer days around the holidays.

Reference breakdown and 1:1 scale



Reference image

The alley is about 5 m wall to wall and 20 m tall, framed by two mill buildings and capped by a skybridge 6 m up. All sizes below are estimates read from the photo using known human-scale anchors: the handrail (100 cm), stair risers (18 cm), door heights (210 to 260 cm) and Victorian brick coursing. Lock them on Day 3 by camera-matching the photo in Blender and checking each one against the blockout.

Scene zones (front to back)

1. Foreground: debris-covered ground, planks, plywood sheet, cables, moss patches, dead branches.
2. Left wall: dark sooty brick mill, tall blind arch recesses at ground level, pilasters, drainpipe, ivy falling from the parapet, glazed upper windows.
3. Right wall: grey painted brick building, tall steel-framed multi-pane windows, stone sills and lintels, a projecting mossy ledge, a basement light well with steps and a green tube handrail.
4. Midground: two-tier skybridge (timber-clad enclosed bridge over a steel girder deck), low arched loading door with rusted sheet doors on the left.
5. Background: red brick end building, tall square chimney, roof overgrowth, bright overgrown opening at the alley's far end (the focal light).

Measurements

Element	Width (cm)	Height (cm)	Depth (cm)	Notes
Alley, wall to wall	500		4000 visible length	about 400 clear

Element	Width (cm)	Height (cm)	Depth (cm)	Notes
				walkway beside the light well
Left mill height		2000		5 storeys at 400 floor to floor, plus parapet
Right building height		1900		5 storeys at about 380 floor to floor
Brick (Victorian imperial)	23	7.6	11	1 cm joints, so 8.6 cm per course, about 12 courses per metre
Left pilaster	60	full height	25 projection	every 400 along the wall
Left blind arch recess	200	550	25 recess	round head, bricked-up infill
Left arched loading door	240	320	40 reveal	rusted sheet doors
Right window opening	140	220	30 reveal	steel frame, panes about 25 x 35
Right window bay spacing	300	380		160 of brick pier between openings
Stone sill	160	15	20	5 projection, drip groove
Stone lintel	170	30	30	flush or 2 projection
Right mossy ledge / plinth	runs wall	50	60	top surface takes world-aligned moss
Basement light well	120	200 drop	380	11 risers at 18, treads 28
Handrail	4.2 diameter	100		posts every 150, painted green
Skybridge, enclosed tier	500 span	300	250	underside at about 600 above ground
Skybridge, girder deck	500 span	40	250	steel I-beams about 30 deep, sits under the enclosed tier
Chimney stack	180	2800 to top	180	top at about 28 m, corbelled cap
Cast-iron drainpipe	10 diameter	full height		hopper head 30 x 30, brackets every 180
Plywood sheet	122	1.8	244	prop
Timber plank	15	5	240	prop
Ground setts (cobbles)	10		20	mostly buried under mud and debris

Metrics to hold for the whole project

- Grid snap 25 cm; modules on 100, 200 and 400 cm widths.
- Floor to floor 400 (left) and 380 (right); pivots at bottom-left-back of each module.
- Texel density 10.24 px/cm (1024 px/m) for surfaces within 5 m of camera; 5.12 px/cm for upper floors and background.
- Brick tiling texture covers 200 x 200 cm with 24 courses (8.33 cm each, a small cheat on the real 8.6) so coursing lines up at every 100 cm module edge.

Modular kit

The kit is 31 pieces in four families that build both facades, the ground and the structural details. Every piece snaps on a 25 cm grid, uses tiling materials or one of three trim sheets, and takes vertex paint for moss, grime and wet.

Rules

- Naming: `SM_MillAlley_<Family>_<Piece>_<W>x<H>`, for example `SM_MillAlley_A_WallFlat_400x400`.
- Pivot at bottom-left-back; walls 400 cm tall so one module = one storey.
- Wall faces use tiling materials (UV0) at the locked texel density; a second UV for lightmaps is optional since Lumen is on.
- Edges get a 1 to 2 cm bevel with weighted normals, so silhouettes catch light without bakes.
- Each family is tested in a kit-bash scene before it goes into the alley.

Family A: left mill (dark sooty brick)

ID	Piece	Size (cm)	Material
A01	Wall flat	400 x 400, 200 x 400, 100 x 400	Brick sooty
A02	Pilaster	60 x 400, 25 projection	Brick sooty
A03	Blind arch recess with brick infill	300 x 600	Brick sooty + infill variant
A04	Upper window, round head, glazed	300 x 400	Brick + metal trim + dirty glass
A05	Arched loading door surround	300 x 400	Brick + stone trim
A06	Outer and inner corner	50 x 400	Brick sooty
A07	String course	400 long	Stone trim
A08	Parapet with coping	400 x 100	Brick + stone trim
A09	Base course / plinth	400 x 60	Brick sooty, wet blend

Family B: right building (grey painted brick)

ID	Piece	Size (cm)	Material
B01	Wall flat, painted	400 x 380, 200 x 380, 100 x 380	Brick painted grey
B02	Window bay (opening, sill, lintel)	300 x 380	Brick painted + stone trim
B03	Steel window insert: intact, broken, gutted	140 x 220	Metal trim + dirty glass
B04	Door bay with timber door	300 x 380	Brick painted + timber trim
B05	Projecting ledge / plinth	400 x 50, 60 deep	Stone trim + world-aligned moss
B06	Light well: walls, steps, landing	120 x 380, 200 drop	Concrete + stone trim
B07	Floor band	400 long	Stone trim
B08	Outer corner	50 x 380	Brick painted
B09	Roof edge with gutter	400 long	Stone + metal trim

Family C: structure and metal

ID	Piece	Size (cm)	Material
C01	Steel I-beam	100, 200, 400 long, 30 deep	Metal trim, rusted paint
C02	Drainpipe: straight, elbow, offset, shoe	100, 200, 400; 10 diameter	Metal trim

ID	Piece	Size (cm)	Material
C03	Hopper head	30 x 30	Metal trim
C04	Pipe and cable bracket, wall hook	10 to 30	Metal trim
C05	Handrail: straight, post, bend, end cap	100, 150, 200	Metal trim, green paint
C06	Gutter run and bracket	200, 400	Metal trim

Family D: ground

ID	Piece	Size (cm)	Material
D01	Ground tile, setts under mud	400 x 400	Ground blend: setts, mud, moss, puddle
D02	Kerb / wall-base strip	400 long	Stone trim
D03	Drain grate and channel	100 x 30	Metal trim
D04	Rubble edge strip (ground meets wall)	400 long	Ground + brick rubble
D05	Broken ground patch (exposed setts)	200 x 200	Ground setts
D06	Puddle insert (shallow depression)	150 x 100	Ground + wet
D07	Wall-base dirt pile	200 long	Ground mud

Trim sheets (2048 x 2048, 10.24 px/cm)

- T01 Stone: sills, lintels, coping, string courses, kerbs.
- T02 Metal: window frames, beams, rails, pipes, gutters, brackets.
- T03 Timber: boards, door planks, skybridge cladding strips, window mullions.

Unique hero features

Seven unique features break the kit's rhythm and give the eye somewhere to land; the skybridge is the centrepiece and gets the most time (Week 6).

#	Feature	Why it matters in the shot	Build approach
H1	Two-tier skybridge	Frames the vanishing point; the only warm colour in the scene	Blender sculpt of rotted cladding; Marmoset bake and texture (peeling red and green paint) to a unique 4K set, girders from C01
H2	Chimney stack and red end building	Silhouette anchor against the bright sky	Mid-poly, tiling red brick with a unique corbelled cap; Nanite
H3	Basement light well with green handrail	Foreground depth cue and leading line	Built from B06 and C05, hand-placed moss, wet steps, debris on landing
H4	Cascading ivy and wall growth	Breaks hard brick edges on the left; adds a second green	Spline or hand-placed ivy cards from the parapet down the drainpipe; buddleia tufts in mortar joints
H5	Overgrown far opening	The light source and focal point at the end of the alley	Dense foliage, bright exposure, volumetric fog shaft
H6	Rusted arched loading doors	Mid-left story beat; texture contrast to brick	Unique doors (sheet steel, hinges, rust bleed) sitting in A05
H7	Broken industrial glazing	Repeats down the right wall;	Three B03 variants plus glass

#	Feature	Why it matters in the shot	Build approach
		catches sky reflections	shard decals and scatter on sills

Full asset list

The scene needs 79 assets: 31 kit pieces, 7 hero features, 22 props, 10 foliage assets and 9 decal sets. Budgets are LOD0 triangles for a game-ready version; keep them even with Nanite on, since reviewers check wireframes. P1 = needed for the reference shot, P2 = needed for alternate angles, P3 = only if time allows.

Kit and heroes

ID	Asset	Count	Tris each	Textures	Priority
A01 to A09	Left mill kit	9	300 to 3,000	Tiling + T01	P1
B01 to B09	Right building kit	9	300 to 4,000	Tiling + T01, T02, T03	P1
C01 to C06	Structure and metal kit	6	100 to 1,500	T02	P1
D01 to D07	Ground kit	7	500 to 4,000	Ground blend	P1
H1	Skybridge	1	40,000	Unique 4K + T02, T03	P1
H2	Chimney and end building	1	8,000	Tiling red brick + unique 2K cap	P1
H3	Light well vignette	1	built from kit	Kit + decals	P1
H4	Ivy and wall growth set	1	15,000 per cluster	Foliage atlas	P1
H5	Far opening foliage mass	1	30,000	Foliage atlas	P1
H6	Arched loading doors	1	3,000	Unique 2K	P2
H7	Broken glazing variants	3	800	T02 + glass	P1

Props and debris

ID	Asset	Size (cm)	Tris	Textures	Priority
P01	Plywood sheet, warped and delaminating	122 x 244	400	Unique 2K	P1
P02	Timber planks, 4 variants	15 x 240	200 each	T03	P1
P03	Broken timber offcuts, 6 variants	10 to 60	100 each	T03	P1
P04	Loose bricks and half bricks, 6 variants	23 x 11	150 each	Unique 1K atlas	P1
P05	Rubble pile cluster, 3 variants	100 to 200	2,500	Ground + brick atlas	P1
P06	Rusted pipe lengths, 3 variants	100 to 300	300	T02	P1
P07	Hanging and loose cables, spline	any	50 per metre	Tiling cable	P1
P08	Fallen gutter section	200	600	T02	P2
P09	Broken window	140 x 220	800	T02	P2

ID	Asset	Size (cm)	Tris	Textures	Priority
	frame on ground				
P10	Glass shards scatter, 8 pieces	5 to 20	20 each	Glass	P2
P11	Dead branches, 5 variants	50 to 200	400	Foliage atlas	P1
P12	Corrugated sheet, bent	80 x 200	1,200	Unique 1K	P2
P13	Rusted barrel or drum, on its side	60 diameter	1,500	Unique 2K	P3
P14	Crate, broken	60 x 60	600	T03	P3
P15	Plastic litter: bags, bottles, cans	5 to 40	100 each	Unique 1K atlas	P2
P16	Paper and cardboard scraps	10 to 50	20 each	Atlas	P2
P17	Wall-mounted junction box	30 x 40	400	T02	P2
P18	Old wall lamp bracket	40	600	T02	P2
P19	Fallen roof slates	25 x 50	20 each	Atlas	P3
P20	Pallet, rotten	120 x 100	800	T03	P3
P21	Wire mesh panel, crumpled	100 x 200	1,000	Opacity mask	P3
P22	Hose or flex pipe, spline	any	60 per metre	Tiling rubber	P3

Foliage

ID	Asset	Tris	Priority
F01	Ivy cards and vine splines	200 per card	P1
F02	Moss clumps, 3 sizes	50 to 200	P1
F03	Grass and weed tufts, 6 variants	300 to 800	P1
F04	Buddleia bush (grows from walls)	4,000	P1
F05	Dead leaves scatter	4 per leaf	P1
F06	Brambles	3,000	P2
F07	Small birch and sycamore saplings	8,000	P1
F08	Background trees (far opening)	20,000	P1
F09	Roof-edge weeds for the chimney building	1,500	P2
F10	Ferns in damp corners	1,200	P3

Decal sets

ID	Decal set	Priority
X01	Water leaks and streaks under sills and pipes	P1
X02	Soot and grime gradients up from the ground	P1
X03	Moss and algae patches	P1

ID	Decal set	Priority
X04	Brick damage, spalling and cracks	P1
X05	Peeling paint edges (right wall)	P1
X06	Rust bleed from metal fixings	P1
X07	Puddle and wet-ground patches	P1
X08	Oil stains and mud tracks	P2
X09	Old painted signage, faded	P3

Materials and texture plan

Ten tiling materials, three trim sheets and one master material carry almost every surface; unique texture sets are kept to the skybridge, doors and a few props.

Tiling materials (CC0 or procedural bases customized in Blender with custom edge-wear and leak masks, 2K, height-blend ready)

ID	Material	Tile size (cm)	Used on
M01	Brick, sooty dark (Victorian, recessed mortar)	200 x 200	Family A
M02	Brick, grey limewash peeling to red brick	200 x 200	Family B
M03	Brick, red clean (background)	200 x 200	H2
M04	Brick infill, mismatched	200 x 200	A03
M05	Ground setts under mud	200 x 200	D01, D05
M06	Mud and dirt with organic debris	200 x 200	Ground blend layer
M07	Concrete, weathered	200 x 200	B06 light well
M08	Moss and algae	100 x 100	Blend layer, world-aligned on top faces
M09	Timber, painted and peeling (red, green)	200 x 200	H1, doors
M10	Rusted painted metal	100 x 100	Large metal surfaces

Master materials (Unreal)

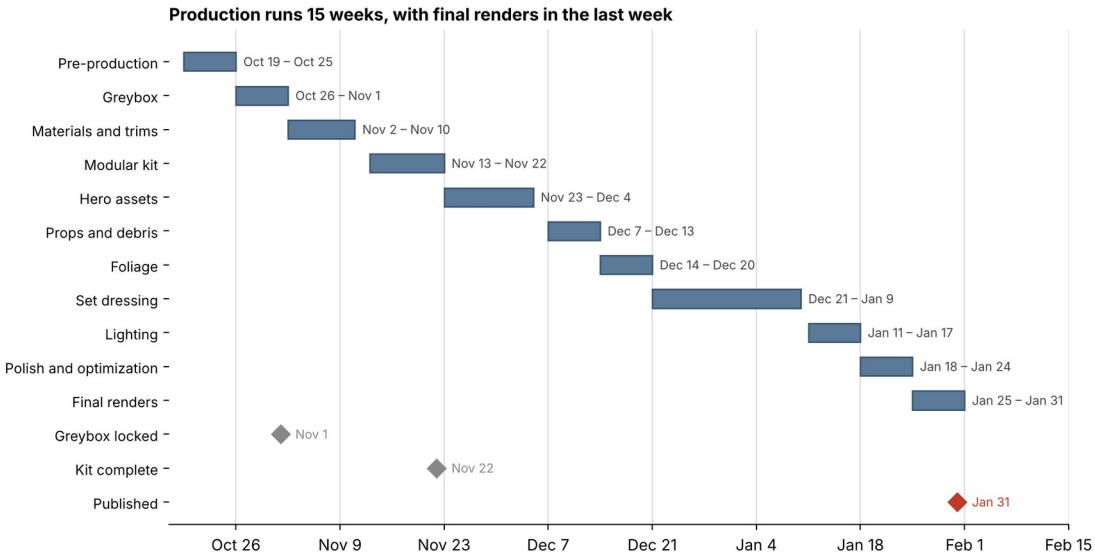
- `M_Master_Kit`: tiling base + 3-layer vertex paint height blend (R moss, G grime, B wet), world-aligned moss on up-facing surfaces, macro variation to kill tiling, edge wear from a baked mask, and world-space leak streaks under sills and ledges.
- `M_Master_Trim`: trim sheet base with the same vertex blend.
- `M_Master_Decal`: DBuffer decals with normal and roughness.
- `M_Master_Foliage`: two-sided, subsurface, wind.
- `M_Master_Glass`: dirty, cracked, roughness from a grime mask.

Unique texture sets (Marmoset Toolbag texturing)

- H1 skybridge: 4K, 2 sets (cladding, structure).
- H2 chimney cap: 2K. H6 loading doors: 2K. P01 plywood: 2K.
- Prop atlases: bricks, litter, paper, slates at 1K to 2K each.

Daily schedule

75 workdays from Oct 19, 2026 to Jan 31, 2027, on a Monday, Tuesday, Friday, Saturday, Sunday week with Wednesday and Thursday off: 68 production days and 7 buffer days. Each week ends with a review against the reference; tick each day off as you go.



Production timeline

The kit is done by Sun Nov 22, so everything after it builds on locked metrics; set dressing absorbs the two holiday weeks.

Week 1 · Oct 19 to 25 · Pre-production

- ☐ **Mon Oct 19** · Build a reference board (a Blender reference scene or any image board): the photo plus 40+ supporting shots (northern English mills, Victorian brick, steel windows, skybridges, ivy on brick). *Done when:* sorted by scene zone.
- ☐ **Tue Oct 20** · Paint over the reference: circle every repeating element (kit) and every one-off (hero); draft kit IDs. *Done when:* annotated paintover saved.
- ☐ **Fri Oct 23** · Camera match in Blender: photo as camera background, align the vanishing lines, set focal length; block primitive volumes at the measured sizes. *Done when:* overlay lines up with the photo.
- ☐ **Sat Oct 24** · Lock metrics, naming and folders; set up the UE5 project and source control (Git LFS or Perforce). *Done when:* metrics sheet and empty project committed.
- ☐ **Sun Oct 25** · Lighting and mood paintover (overcast, cool, wet); Review 1. *Done when:* target image saved next to the reference.

Week 2 · Oct 26 to Nov 1 · Greybox

- ☐ **Mon Oct 26** · Greybox ground and both building masses in UE5 at scale; place a 180 cm mannequin. *Done when:* walkable alley.
- ☐ **Tue Oct 27** · Greybox the left facade with placeholder A01 to A09 at final dimensions. *Done when:* left wall reads like the photo.
- ☐ **Fri Oct 30** · Greybox the right facade with B01 to B09, including the ledge and light well. *Done when:* right wall reads like the photo.
- ☐ **Sat Oct 31** · Greybox skybridge, chimney, end building and loading door; set Camera 1 (reference match) and 3 alternate cameras. *Done when:* full silhouette matches.

- ☐ **Sun Nov 1** · Blockout lighting (sun, sky, fog), walkthrough, Review 2; lock the greybox. *Done when:* side-by-side greybox vs reference.

Week 3 · Nov 2 to 8 · Tiling materials

- ☐ **Mon Nov 2** · Source CC0 base materials (ambientCG, Poly Haven, Blendkit free) for sooty brick, painted brick, setts, mud, concrete, moss, timber and rusted metal; rescale each to the 200 cm tile metric. *Done when:* 10 base sets in a Blender material library.
- ☐ **Tue Nov 3** · Build custom mask node groups in Blender: edge wear (Bevel and AO nodes into a noise-broken threshold) and leak streaks (object-space vertical gradient times stretched noise, starting under sills and ledges). *Done when:* both masks drive a test wall module.
- ☐ **Fri Nov 6** · Brick sets M01 to M04: soot, limewash peel and infill variations layered with the mask groups; bake to 2K tiling sets. *Done when:* brick sets tile cleanly at 200 cm.
- ☐ **Sat Nov 7** · M05 to M10 (setts, mud, concrete, moss, timber, rusted metal): adjust and bake to 2K; bake edge-wear and leak masks for the kit test pieces. *Done when:* all 10 materials exported.
- ☐ **Sun Nov 8** · Build `M_Master_Kit` in UE5: vertex height blend, world-aligned moss, a world-space leak streak and a baked edge-wear input; Review 3. *Done when:* painting moss, grime and wet works.

Week 4 · Nov 9 to 15 · Trim sheets and Family A

- ☐ **Mon Nov 9** · Model T01 stone and T02 metal trim strips in Blender; bake to 2048 sheets in Marmoset. *Done when:* 2048 sheets with normal, AO, height.
- ☐ **Tue Nov 10** · T03 timber trim sheet; `M_Master_Trim`; set up a texel density checker. *Done when:* all three sheets in engine.
- ☐ **Fri Nov 13** · Model A01, A02, A06, A09 with bevels and UVs. *Done when:* snap test passes on the 25 cm grid.
- ☐ **Sat Nov 14** · Model A03 blind arch with infill, A05 door surround, A07 string course, A08 parapet. *Done when:* in engine with materials.
- ☐ **Sun Nov 15** · Model A04 window; replace the left greybox with the kit; Review 4. *Done when:* left wall fully kit-built.

Week 5 · Nov 16 to 22 · Families B, C and D

- ☐ **Mon Nov 16** · Model B01, B07, B08, B09. *Done when:* snap test passes.
- ☐ **Tue Nov 17** · Model B02 window bay and the three B03 steel window variants. *Done when:* bays swap variants cleanly.
- ☐ **Fri Nov 20** · Model B04 door bay, B05 ledge, B06 light well. *Done when:* right wall fully kit-built.
- ☐ **Sat Nov 21** · Model Family C: beams, drainpipes, hopper head, brackets, handrail, gutters. *Done when:* pipes run full height with no gaps.
- ☐ **Sun Nov 22** · Model Family D ground pieces; replace all remaining greybox; Review 5 with a kit-only screenshot. *Done when:* the alley is 100% kit.

Week 6 · Nov 23 to 29 · Hero: skybridge (H1)

- ☐ **Mon Nov 23** · Plan the skybridge from reference; model the base mesh (frame, cladding boards, windows, girder deck). *Done when:* matches greybox volume.
- ☐ **Tue Nov 24** · Sculpt in Blender: warped and broken boards, rot, sagging, splintered window frames. *Done when:* high poly reads at Camera 1 distance.
- ☐ **Fri Nov 27** · Retopo the low poly, UV into two 4K sets. *Done when:* under 40,000 tris, clean UVs.
- ☐ **Sat Nov 28** · Bake in Marmoset; start the Marmoset texture project: peeling red and green paint, rot, rust on girders. *Done when:* base materials and masks in.
- ☐ **Sun Nov 29** · Finish texturing, import, place; Review 6 against the reference. *Done when:* skybridge final in Camera 1.

Week 7 · Nov 30 to Dec 6 · Heroes H2, H3, H6, H7

- ☐ **Mon Nov 30** · Model the chimney and red end building; sculpt the corbelled cap. *Done when:* silhouette locked.
- ☐ **Tue Dec 1** · Texture the chimney cap; model and texture H6 rusted loading doors. *Done when:* both final in engine.
- ☐ **Fri Dec 4** · H7 broken glazing variants and `M_Master_Glass`; assemble H3 light well vignette; Review 7. *Done when:* all heroes in scene.
- ☐ **Sat Dec 5** · Buffer: catch up on anything late from Weeks 3 to 7.
- ☐ **Sun Dec 6** · Buffer.

Week 8 · Dec 7 to 13 · Props and debris

- ☐ **Mon Dec 7** · P01 plywood, P02 planks, P03 offcuts. *Done when:* textured and placed in a prop gallery level.
- ☐ **Tue Dec 8** · P04 bricks and P05 rubble piles (sculpt, bake, atlas). *Done when:* 3 rubble clusters ready.
- ☐ **Fri Dec 11** · P06 pipes, P07 cable spline, P08 fallen gutter, P09 broken frame. *Done when:* cable spline usable in level.
- ☐ **Sat Dec 12** · P10 glass shards, P11 dead branches, P12 corrugated sheet, P15 litter, P16 paper. *Done when:* all P1 and P2 small props done.
- ☐ **Sun Dec 13** · P17 junction box, P18 lamp bracket, then P3 props if time; Review 8 with a prop sheet. *Done when:* prop gallery screenshot.

Week 9 · Dec 14 to 20 · Foliage

- ☐ **Mon Dec 14** · Model leaves (ivy, buddleia, grass, weeds) in Blender and bake a foliage atlas in Marmoset. *Done when:* atlas with opacity, normal, translucency.
- ☐ **Tue Dec 15** · F01 ivy cards and vine splines; place H4 ivy down the left wall and drainpipe. *Done when:* ivy breaks the parapet line like the photo.
- ☐ **Fri Dec 18** · F02 moss clumps, F03 grass and weeds, F05 dead leaves. *Done when:* paintable with the foliage tool.
- ☐ **Sat Dec 19** · F04 buddleia, F06 brambles, F07 saplings. *Done when:* all with wind.
- ☐ **Sun Dec 20** · F08 background trees and the H5 far opening mass; Review 9. *Done when:* far end reads bright and green.

Week 10 · Dec 21 to 27 · Decals and vertex paint

- ☐ **Mon Dec 21** · Build decals X01 to X04 and `M_Master_Decal`. *Done when:* decals project cleanly on brick.
- ☐ **Tue Dec 22** · Build decals X05 to X08. *Done when:* full decal library in engine.
- ☐ **Fri Dec 25** · Buffer (Christmas).
- ☐ **Sat Dec 26** · Vertex paint both facades: moss where water sits, grime rising from the ground, wet under leaks. *Done when:* no visible tiling in Camera 1.
- ☐ **Sun Dec 27** · Place decals: leaks under sills, rust bleed under fixings, soot gradients, peeling edges. Review 10: Camera 1 vs the photo; list fixes. *Done when:* every sill and pipe has a story and the fix list is written.

Week 11 · Dec 28 to Jan 3 · Set dressing

- ☐ **Mon Dec 28** · Ground dressing: rubble edges along wall bases, debris following drainage and foot traffic. *Done when:* ground matches photo density.
- ☐ **Tue Dec 29** · Place P1 props in Camera 1 for composition (plywood, planks, branches, cables). *Done when:* leading lines point to the far opening.
- ☐ **Fri Jan 1** · Buffer (New Year's Day).

- ☐ **Sat Jan 2** · Foliage placement: weeds in cracks, moss on ledges and steps, buddleia in mortar; Review 11. *Done when:* foreground to background reads clearly.
- ☐ **Sun Jan 3** · Buffer.

Week 12 · Jan 4 to 10 · Dressing for alternate angles

- ☐ **Mon Jan 4** · Dress the alternate cameras with P2 props, glass, litter, paper. *Done when:* no empty corners in any camera.
- ☐ **Tue Jan 5** · Puddles and wetness: X07 decals, D06 inserts, wet roughness on the ground. *Done when:* puddles reflect the sky.
- ☐ **Fri Jan 8** · Buffer.
- ☐ **Sat Jan 9** · Full walkthrough: fix floating objects, seams, repeated props; Review 12. *Done when:* clean walkthrough capture.
- ☐ **Sun Jan 10** · Buffer.

Week 13 · Jan 11 to 17 · Lighting and atmosphere

- ☐ **Mon Jan 11** · Overcast sky (HDRI or Sky Atmosphere), soft directional, skylight, Lumen settings; lock exposure. *Done when:* flat, cool base matches the photo.
- ☐ **Tue Jan 12** · Height fog and volumetrics; a soft light shaft at the far opening; bounce and fill lights. *Done when:* depth reads in three layers.
- ☐ **Fri Jan 15** · Post process: colour grade toward the photo, light vignette, grain; tune reflections on glass and puddles. *Done when:* Camera 1 sits next to the photo convincingly.
- ☐ **Sat Jan 16** · Light the alternate cameras. *Done when:* each has a clear focal point.
- ☐ **Sun Jan 17** · Review 13: share lit renders with peers or a mentor for critique. *Done when:* critique notes collected.

Week 14 · Jan 18 to 24 · Polish and optimization

- ☐ **Mon Jan 18** · Apply critique notes. *Done when:* every note fixed or consciously skipped.
- ☐ **Tue Jan 19** · Material polish: tiling breakup, roughness ranges, greyscale value check. *Done when:* scene reads in greyscale.
- ☐ **Fri Jan 22** · Optimization: Nanite and LODs, texture streaming, collision, profile with stat unit and GPU visualizer. *Done when:* stable frame rate in a walkthrough.
- ☐ **Sat Jan 23** · Audit texel density, wireframes, naming; build a kit breakdown level with every module laid out. *Done when:* breakdown level ready to shoot.
- ☐ **Sun Jan 24** · Final polish in Camera 1; lock the scene; Review 14. *Done when:* scene locked.

Week 15 · Jan 25 to 31 · Final renders

- ☐ **Mon Jan 25** · Beauty renders at 4K with Movie Render Queue: Camera 1 plus 4 to 6 angles. *Done when:* all beauty shots exported.
- ☐ **Tue Jan 26** · Breakdown renders: kit sheet, wireframe, clay, lighting only, material spheres, trim sheets. *Done when:* breakdown set exported.
- ☐ **Fri Jan 29** · Marmoset turntables for H1, H2, H6 and a prop sheet; render the flythrough sequence. *Done when:* all renders on disk.
- ☐ **Sat Jan 30** · Edit the 30 to 45 second flythrough in Blender's Video Sequencer; make the reference vs final side-by-side. *Done when:* video exported.
- ☐ **Sun Jan 31** · Lay out and publish the ArtStation post with breakdown text; update portfolio. *Done when:* project is live.

Free resources

Free CC0 libraries cover the base materials and fill props, so your own time goes into the kit, the heroes and the custom masks. Credit anything you didn't make in the final post.

Resource	Use it for	Cost and license
ambientCG	Base PBR materials: brick, concrete, metal, ground	Free, CC0
Poly Haven	Base materials, overcast HDRIs, some props	Assets free, CC0; its Blender add-on is paid, so download from the site
Blendkit (formerly BlenderKit)	Materials and fill props from inside Blender	60,000+ free assets, royalty-free or CC0; paid tier for the full library
Fab	Megascans and environment packs for UE5	Some assets free, most paid since 2025; Megascans claimed before 2025 stay yours
Texel Density Checker	Hitting 10.24 px/cm on every module	Free extension, Blender 4.2+
Cell Fracture	Broken bricks, rubble, smashed boards	Free extension, Blender 4.2+
Ivy Gen	Ivy cascading from the parapet (H4)	Free extension
Node Wrangler	Fast material and mask node setups	Free, ships with Blender
Send to Unreal	One-click Blender to UE5 export	Free; check it supports your Blender version, plain FBX export works too

Sources: Poly Haven add-on, Megascans on Fab, Blendkit free assets, free texture libraries.

Final renders and presentation

The ArtStation post leads with the reference-matched beauty shot, then proves the kit and the heroes in that order.

Render list (post order)

1. Hero beauty shot: Camera 1, matched to the photo, 4K.
2. Reference vs final side-by-side.
3. Four to six alternate angles: reverse shot from the far opening, low angle up at the skybridge, light well close-up, ivy and drainpipe detail, a top-down overview.
4. Modular kit sheet: every A to D piece laid out with IDs, plus a few walls assembled two different ways from the same pieces.
5. Kit assembly GIF: the right facade building up module by module.
6. Trim sheet and tiling material sheet: flat textures plus material spheres.
7. Hero turntables: skybridge, chimney cap, loading doors (Marmoset).
8. Wireframe and clay overlays of Camera 1.
9. Vertex paint and decal breakdown: same wall, with and without.
10. Lighting-only pass.
11. Flythrough video, 30 to 45 seconds.

Post text to write

- One line on the goal (a 1:1 study of the reference built on a AAA modular pipeline) and the time frame.
- Kit stats: number of modules, trim sheets, tiling materials.
- Tools used and what each did.
- Credit for the reference photo and for every third-party material or prop, with its source.